

ECG & HEART SOUNDS

FINAL REPORT

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1 Abstract

The synchronization between the electrical excitation and mechanical contraction of the heart is fundamental to cardiovascular physiology. This study aimed to record and analyze electrocardiogram (ECG) and phonocardiogram (PCG) signals under varying physiological conditions, while evaluating the efficacy of digital signal processing techniques in enhancing diagnostic utility. Experimental data were acquired from a healthy subject using a BIOPAC system under conditions of rest, postural changes, and physical exertion. Raw signals were processed using a linear-phase FIR band-pass filter (0.5–50 Hz) to attenuate noise and baseline wander, followed by the DF1 algorithm for QRS detection and a Shannon energy-based method for heart sound identification. The results demonstrated that filtering and signal averaging significantly improved the Signal-to-Noise Ratio (SNR), increasing it from 55.43 in a single beat to 67.71 in the averaged signal. Physiologically, heart rate increased during standing and deep breathing, although the difference in the latter was not statistically significant ($p > 0.05$). Furthermore, post-exercise analysis revealed a 49% increase in heart rate accompanied by a marked shortening of the diastolic interval, confirming strong electro-mechanical coupling. These findings underscore the critical role of advanced signal processing in extracting reliable clinical data from noisy physiological recordings and characterizing cardiac dynamics.

2 Introduction

The heart's mechanical contraction is driven by an intrinsic electrical conduction system. This process, the anatomical pathway of which is illustrated in Figure 1, begins with an electrical impulse at the Sinoatrial (SA) node, traveling through the atria to the Atrioventricular (AV) node, and continuing through the Bundle of His and Purkinje fibers to the ventricles. This electrical activity is recorded via the Electrocardiogram (ECG). The resulting signal consists of distinct waveforms: the P-wave (atrial depolarization), the QRS complex (ventricular depolarization), and the T-wave (ventricular repolarization).[10]

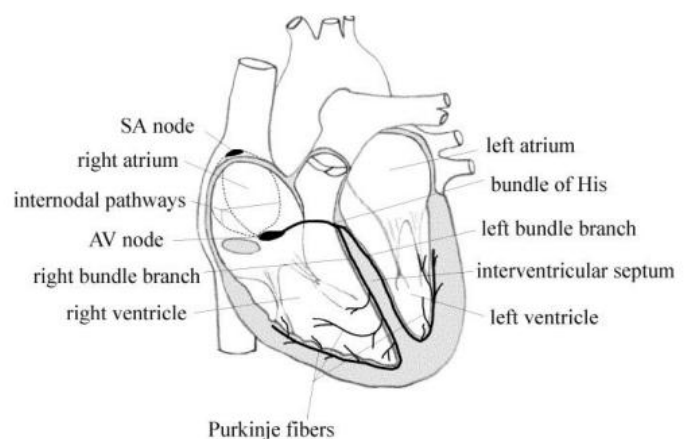


Figure 1: The heart mechanism

From a signal processing perspective, a standard ECG device typically covers a frequency range of 0.05 [Hz] to 150 [Hz], while the QRS complex is characterized by higher frequencies, mainly between 10 [Hz] and 40 [Hz]. To ensure accurate identification of all ECG components, whose standard morphology and segments are illustrated in Figure 2, and avoid aliasing, a phenomenon where high-frequency signals are misinterpreted as lower frequencies due to insufficient sampling, an optimal sampling rate of at least 500-1000 [Hz] is required (following the Nyquist theorem). Common noises in ECG measurements include power line interference (50-60 [Hz]), muscle tremors (EMG), and Baseline Wander, which is a low-frequency drift of the signal's base voltage caused by breathing or electrode movement.[9]

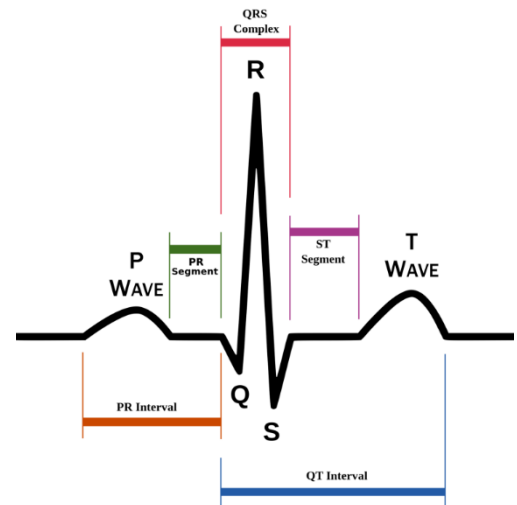


Figure 2: Standard ECG waveform [4]

The cardiac cycle also involves mechanical events. Blood flow is regulated by the opening and closing of valves: the Atrioventricular (AV) valves (mitral and tricuspid) and the Semilunar valves (aortic and pulmonary), which operate under pressure constraints. These mechanical actions produce heart sounds (PCG). The first heart sound (S1 also known as "Lub") occurs at the onset of ventricular systole as AV valves close. It is typically louder, longer, and lower-pitched compared to the second heart sound (S2 also known as "Dup"), which is shorter and higher-pitched, signaling the closure of semilunar valves. S1 is the loudest sound due to the forceful ventricular contraction and the closing of the larger AV valves. These sounds typically occupy a frequency range of 20 [Hz] to 200 [Hz].[8]

There is a clear temporal relationship between electrical and mechanical cycles: S1 aligns with the peak of the R-wave (or slightly after), while S2 occurs at the end of the T-wave. This experiment aims to record and analyze ECG and heart sounds under various physiological conditions (rest and post-exercise) to understand the electro-mechanical synchronization and the cardiac response to physical stress.

3 Materials and methods

3.1 Experimental Procedure

The experiment was conducted on a single healthy female participant. All measurements were performed in the Biomedical Engineering Laboratory using a BIOPAC data acquisition system. ECG and heart sound (HS) signals were sampled at $F_s = 500$ [Hz], a sampling rate sufficient to capture the full spectral content of the ECG signal while avoiding aliasing.[3]

Surface electrodes were attached according to the standard bipolar limb-lead configuration, as illustrated in Figure 3, allowing direct measurement of Leads I and III. Before each recording session, the system was calibrated while the subject remained seated and relaxed to minimize motion and muscle artifacts."

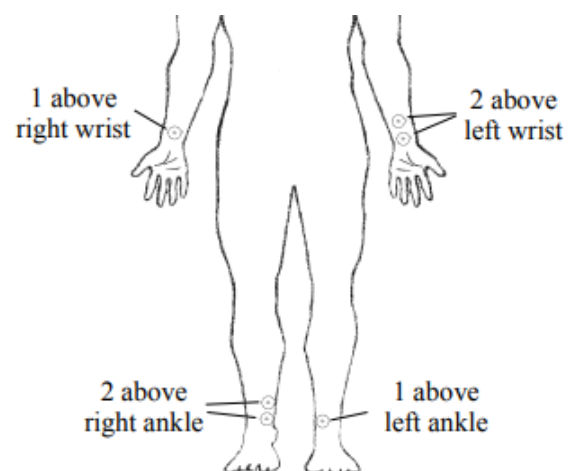


Figure 3: Lead I and Lead III electrode Placement.

The experimental protocol was divided into three stages. In the first stage, ECG signals were recorded under three physiological conditions: seated rest (30 [sec]), standing with normal breathing (10 [sec]), and standing with deep breathing (10 [sec]). Data acquisition was paused during posture transitions to reduce electromyographic interference.

In the second stage, ECG was recorded continuously for one minute while the subject was seated and breathing normally. This segment was used for heart rate variability (HRV) and signal quality analysis.

In the third stage, heart sounds were recorded simultaneously with ECG. A stethoscope was placed at the cardiac apex, specifically in the region between the Tricuspid (T) and Mitral (M) points illustrated in Figure 4, which was identified using anatomical landmarks. ECG electrodes were arranged to record Lead II. Heart sounds and ECG were recorded for 20 [sec] during seated rest. The subject then performed 60 [sec] of physical activity (without recording), followed by an additional 10 [sec] recording in the seated position.

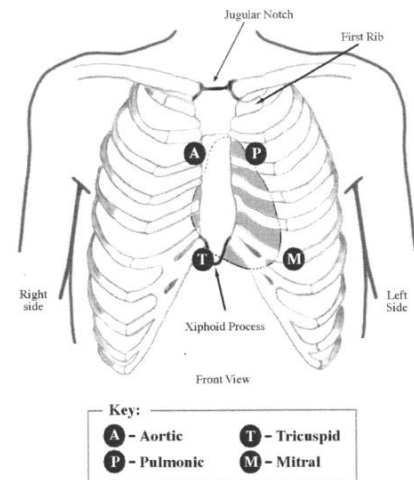


Figure 4: Stethoscope Positions for Optimal Detection of Heart Valve Function

3.2.1 Subject Profile

The subject was a 24-year-old healthy female, 167 [cm] in height and weighing 58 [kg]. All measurements were obtained from the same subject, enabling paired comparisons between physiological conditions.

3.2.2 Einthoven's Law

The relationship between the standard bipolar limb leads is defined by Einthoven's law [7]:

$$(1) \text{ Lead II} = \text{Lead I} + \text{Lead III}$$

This relationship was used to reconstruct Lead II from the measured Leads I and III. Lead II was selected for further analysis because it typically exhibits a high-amplitude and clearly defined QRS complex, making it well suited for R-peak detection.

3.2.3 Signal Filtering

To reduce noise while preserving clinically relevant ECG features, all ECG signals were processed using a linear-phase FIR band-pass filter with cutoff frequencies of 0.5 [Hz] and 50 [Hz] [2,6]. This frequency range effectively attenuates low-frequency baseline wander caused by respiration and electrode motion, as well as high-frequency noise such as muscle activity, while preserving the dominant spectral components of the QRS complex.

An FIR filter was selected due to its linear phase response, which preserves the temporal relationships between ECG components. This property is essential for accurate R-peak detection, heart rate variability analysis, and synchronization between the electrical ECG signal and mechanical heart sounds. Although FIR filters typically require a higher filter order, the computational load was acceptable for offline signal processing in this experiment.

3.2.4 QRS Detection Algorithm

QRS complexes were identified from the filtered Lead II signal using the DF1 detection algorithm, originally proposed by Engelse and Zeelenberg and later evaluated by Friesen et al.[5] This method was selected based on its demonstrated robustness to various noise sources and its favorable trade-off between detection accuracy and computational complexity.

The algorithm consists of the following processing steps:

Differentiation: The ECG signal $X(n)$ was differentiated to enhance rapid slope changes associated with the QRS complex:

$$(2) Y_0(n) = X(n) - X(n - 8)$$

The delay of eight samples produces a notch frequency given by:

$$(3) f = \frac{F_s}{k}$$

where $F_s = 500$ [Hz] and $k = 8$, corresponding to approximately 62.5 Hz.

Smoothing filter : The differentiated signal was smoothed using a low-pass digital filter:

$$(4) Y_1(n) = Y_0(n) + 4Y_0(n - 1) + 6Y_0(n - 2) + 4Y_0(n - 3) + Y_0(n - 4)$$

Threshold-based detection : Two symmetric thresholds of opposite polarity were defined. Based on preliminary inspection of the data, a threshold value of ± 1.7 was selected. When the filtered signal exceeded the threshold, a 160 ms search window was opened.

Classification : Events detected within the search window were classified according to the number of threshold crossings. Patterns consistent with physiological QRS morphology were accepted, whereas events attributed to baseline shifts or noise were rejected. To prevent multiple detections of a single QRS complex, only the first valid detection was retained.

After QRS identification, the precise R-peak location was determined by searching for the maximum amplitude within the corresponding window in the original ECG signal.

The DF1 algorithm is known for its high noise tolerance, particularly in the presence of baseline drift and muscle artifacts [5]. Its main limitation is its reliance on fixed thresholds, which may reduce robustness under highly non-stationary noise conditions.

3.2.5 Heart Rate Calculation

R-peak occurrence times $\{t_1, t_2, \dots, t_N\}$ were used to compute the RR intervals:

$$(5) \Delta RR_i = t_{i+1} - t_i$$

Heart rate for each cardiac cycle was calculated as [7]:

$$(6) HR_i = \frac{60}{\Delta RR_i}$$

The mean heart rate for each condition was obtained by averaging the heart rate values over all detected cycles.

3.2.6 Heart Rate Variability (HRV)

Heart rate variability was quantified using the standard deviation of the heart rate values [7]:

$$(7) \sigma = \sqrt{\frac{1}{M-1} \sum_{i=1}^M (H R_i - \mu)^2}$$

This metric reflects beat-to-beat variability and is commonly used to assess autonomic regulation of heart rate.

3.2.7 Statistical Analysis

To evaluate differences between standing with normal breathing and standing with deep breathing, a paired, two-tailed t-test was applied [1]. This test was appropriate because both conditions were measured in the same subject.

The t-statistic was computed as:

$$(8) t_{obs} = \frac{\bar{d}}{s/\sqrt{n}}$$

where $\bar{d}$ is the mean of the paired differences, s is their standard deviation, and n is the number of paired observations. A significance level of $\alpha = 0.05$ was used. The p-value was calculated as:

$$(9) PV = 2P(t \geq |t_{obs}|)$$

3.2.8 Heart Sound Detection Algorithm

The first (S1) and second (S2) heart sounds were detected using an energy-based method described in the literature [11]. The algorithm was applied separately to recordings obtained at rest and after physical activity.

Preprocessing : The heart sound signal was band-pass filtered to retain frequency components between approximately 25 and 200 [Hz]. The filtered signal $x(t)$ was normalized according to:

$$(10) \tilde{x}(t) = \frac{x(t)}{\max(|x(t)|)}$$

Energy computation : The third-order Shannon energy was calculated in sliding windows:

$$(11) E_{hs} = -\frac{1}{N} \sum_{i=1}^N |\tilde{x}(i)|^3 \log(|\tilde{x}(i)|^3)$$

The energy signal was standardized using its mean and standard deviation:

$$(12) P_{ha}(t) = \frac{E_{hs}(t) - M(E_{hs})}{S(E_{hs})}$$

Segmentation and identification : An adaptive threshold was applied to the energy envelope to define candidate heart sound regions. Within each region, dominant peaks were detected and classified as S1 or S2 based on the physiological rule that the diastolic interval is longer than the systolic interval.

This approach is computationally efficient and effective under moderate noise conditions, though its performance may degrade in recordings with strong artifacts or murmurs.

4. Results

4.1 Part A: ECG Signal Analysis

4.1.1 Time Domain Analysis The raw ECG signals recorded from Leads I, II, and III are presented in Figure 5 (full recording) and Figure 6 (zoomed-in view). Visual inspection of the raw data (Figure 5) revealed the presence of significant noise artifacts. A prominent low-frequency baseline wander was observed, particularly noticeable in Lead II, where the signal drifted significantly from the isoelectric line. Additionally, high-frequency noise (likely power line interference or EMG noise) was superimposed on the signal, resulting in a distorted appearance of the waveform trace. A distinct movement artifact was also noted around the 31-second mark in all three leads. Figure 6, which displays three consecutive beats of the raw signal, further illustrates how the high-frequency noise obscures the finer details of the P-wave and T-wave morphology, although the R-peaks remain distinguishable.

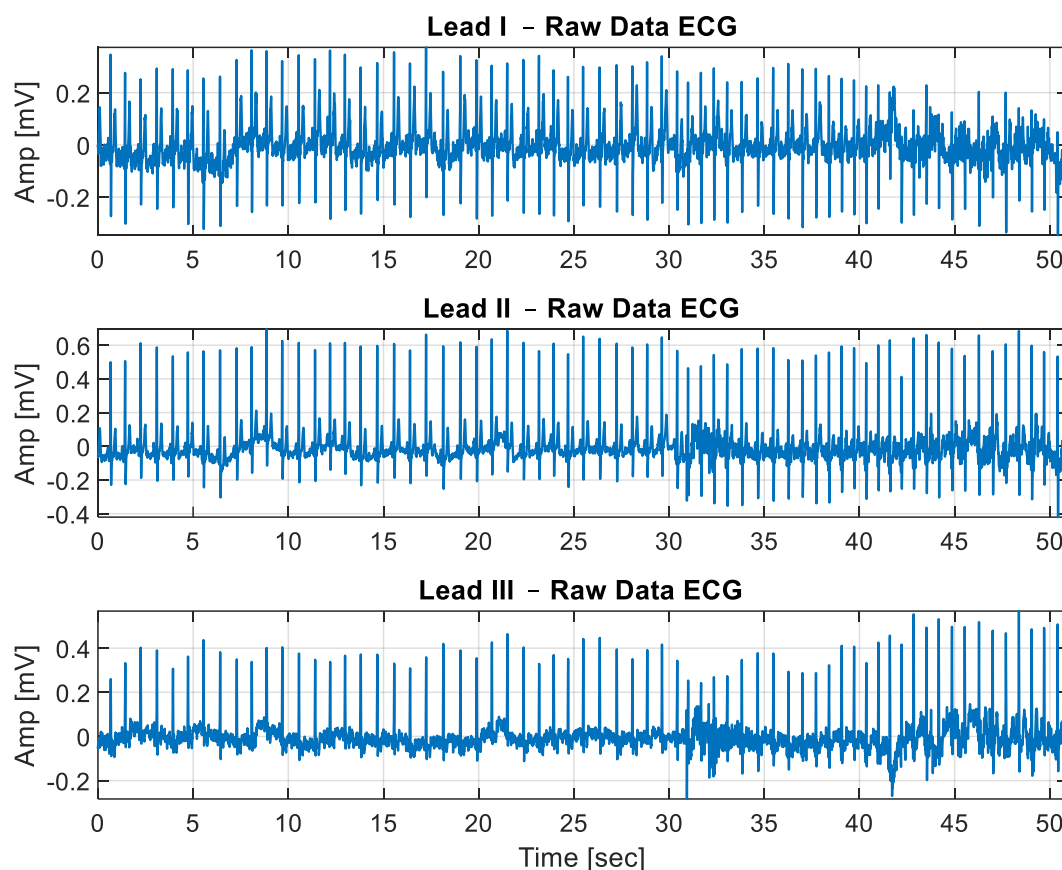


Figure 5: Raw ECG signals (Leads I, II, and III) recorded over a duration of approximately 50 seconds. Note the significant baseline wander and the presence of motion artifacts around $t=31s$.

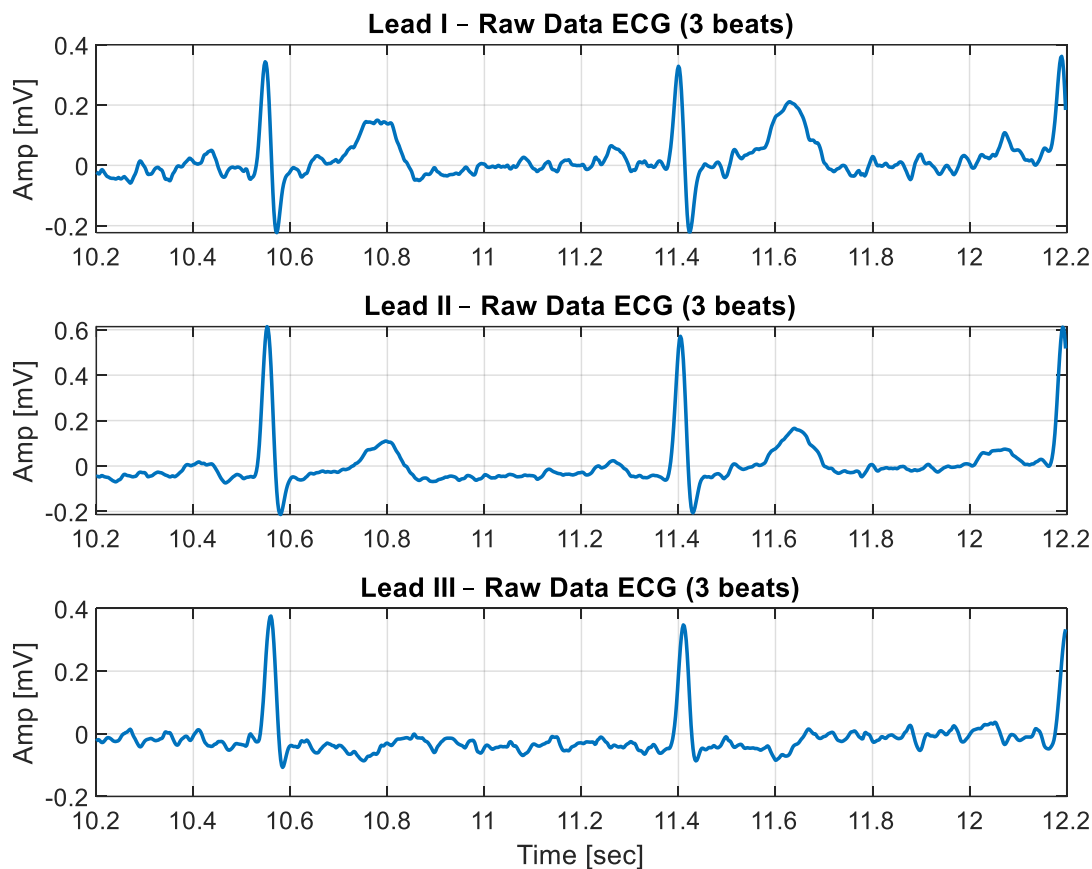


Figure 6: Zoomed-in view of three consecutive heartbeats in the raw signal. The trace appears thick due to high-frequency noise, affecting the clarity of the wave components.

Following the application of the digital filter as described in methods, the signal quality was considerably improved, as shown in Figure 7 and Figure 8. The baseline wander was effectively eliminated, resulting in a stable isoelectric line centered at 0 [mV] across the entire recording duration (Figure 7). While the transient artifact at the 31-second mark remains visible, the low-frequency drift associated with it has been removed. Figure 8 demonstrates the efficacy of the filtration on a finer scale, the high-frequency noise was successfully attenuated, yielding a smooth and clean signal. This enhancement allows for a clear delineation of the P, QRS, and T waves, which is essential for accurate clinical interpretation and further analysis.

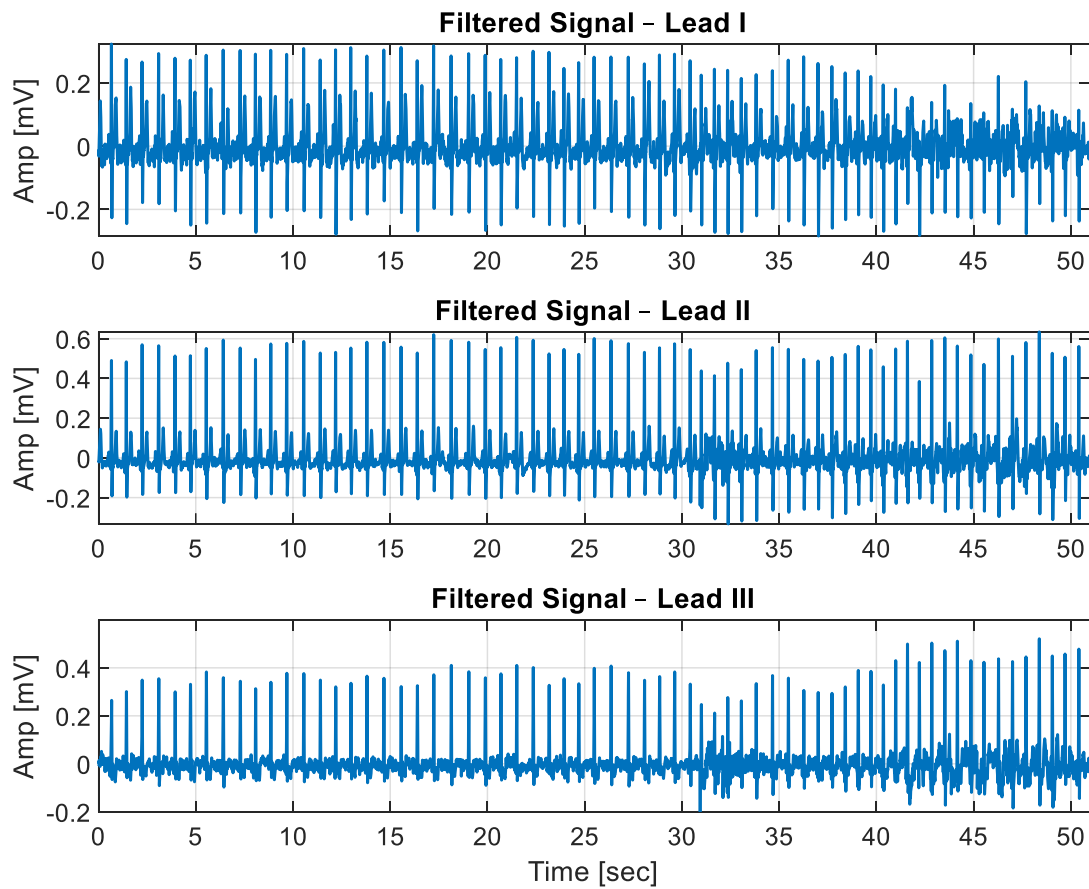


Figure 7: Filtered ECG signals (Leads I, II, and III). The baseline drift has been corrected, providing a stable centered signal.

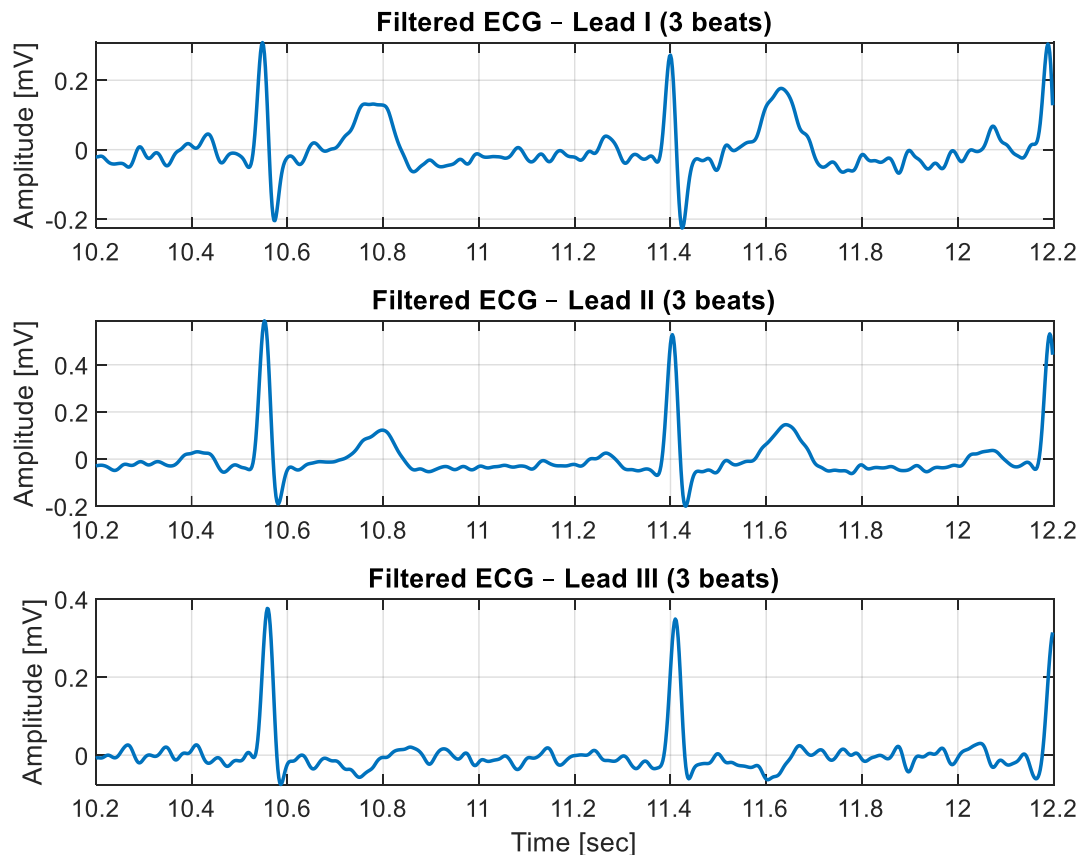


Figure 8: Zoomed-in view of three consecutive heartbeats in the filtered signal. The removal of high-frequency noise reveals a clear morphology of the P, QRS, and T waves.

4.1.2 Frequency Domain Analysis

To further characterize the signal and the effect of the filtration process, a spectral analysis was performed using the Fast Fourier Transform (FFT). Figure 9 displays the normalized magnitude spectrum of the raw ECG signal for Leads I, II, and III. The dominant energy of the ECG signal is concentrated in the lower frequency range (approximately 0.5–40 [Hz]), which contains the essential components of the P, QRS, and T waves.

Following the filtration, the spectral characteristics were re-evaluated, as presented in Figure 10. Concurrently, frequency bands associated with baseline wander (near-DC frequencies) and high-frequency noise were attenuated.

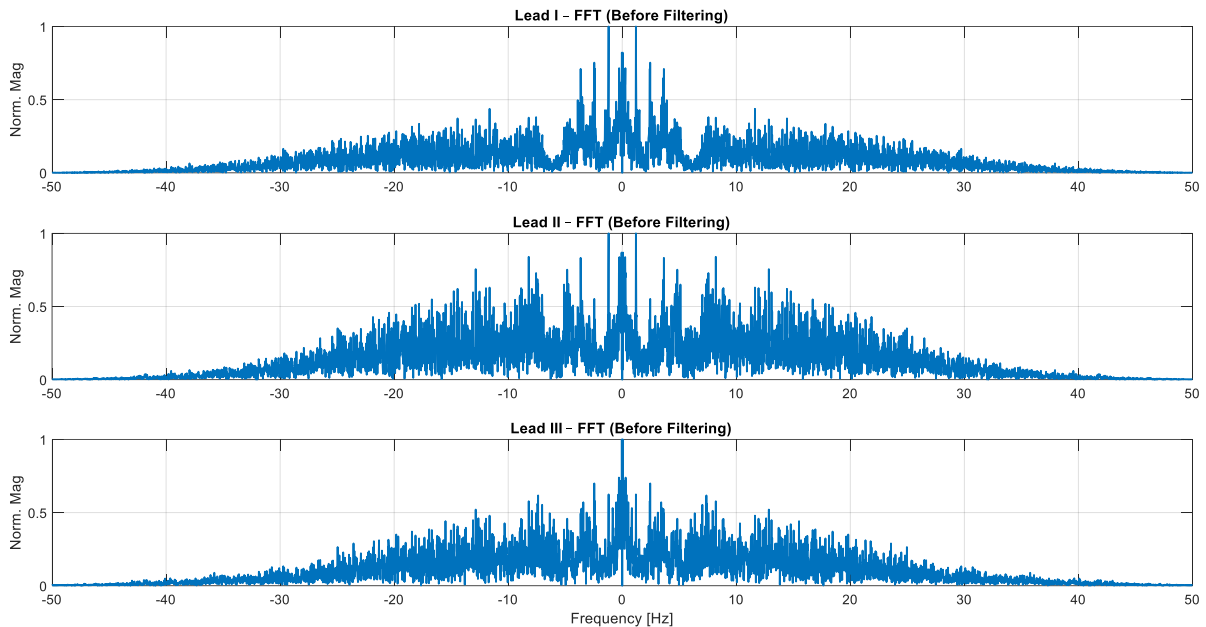


Figure 9: Frequency spectrum (FFT) of the raw ECG signals (Leads I, II, III) prior to filtering.

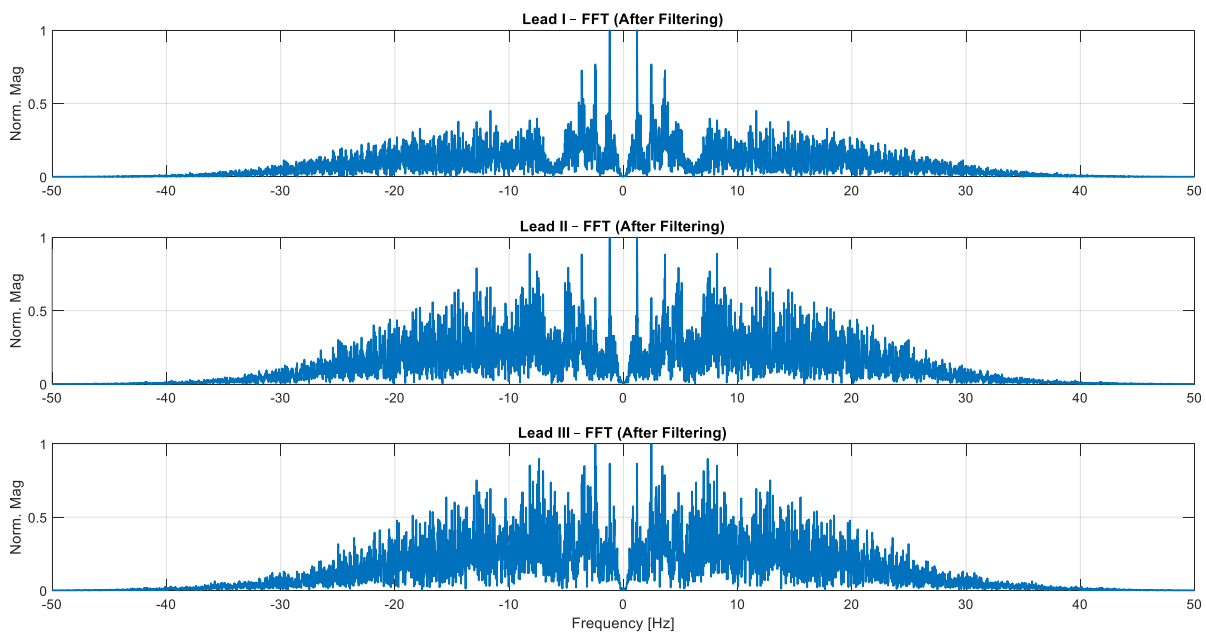


Figure 10: Frequency spectrum (FFT) of the filtered ECG signals (Leads I, II, III).

4.1.3 R-Peak Detection and Heart Rate Analysis

Based on the signal quality assessment in the time domain, Lead II was selected for further processing and heart rate (HR) analysis. An automated algorithm was implemented in MATLAB to detect the R-peaks within the QRS complexes. Figure 11 displays the results of the R-peak detection over the entire duration of the recording.

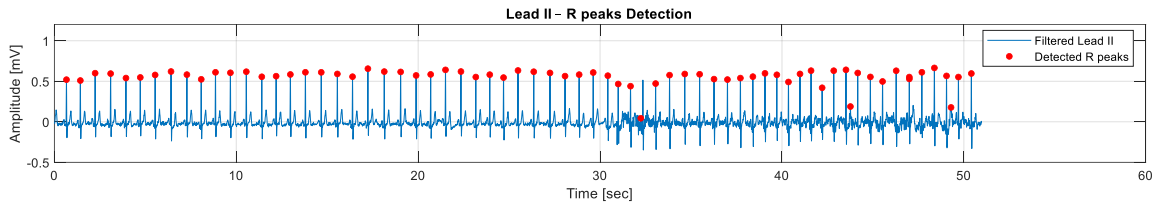


Figure 11: R-peak detection on the filtered Lead II signal over the full recording duration. Red dots indicate the detected R-wave peaks.

The recording was subsequently segmented into three distinct physiological states: resting (seated), standing with normal breathing, and standing with deep breathing. Figure 12 presents the R-peak detection results for each of these states.

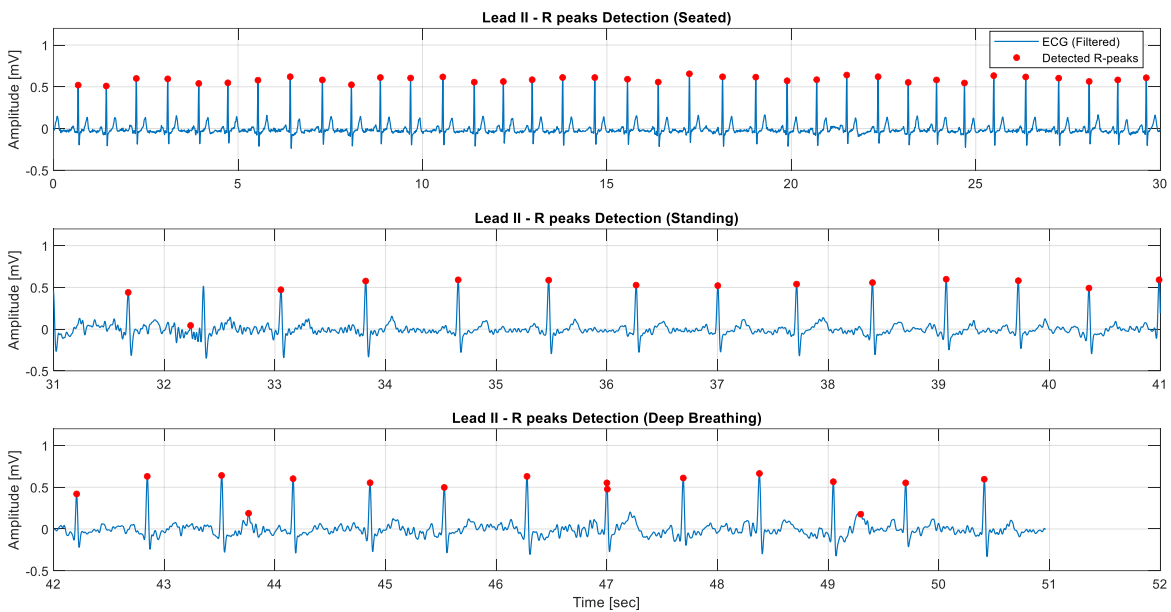


Figure 12: R-peak detection across three physiological states: Seated (top), Standing with normal breathing (middle), and standing with deep breathing (bottom). Note the variation in R-R intervals between the states.

4.1.4 Heart Rate Calculation:

Following the successful detection of R-peaks, the heart rate (HR) statistics were calculated for each physiological state based on the identified cardiac cycles. The mean HR for the resting (seated) position was found to be 72.71 ± 3.7 [BPM], calculated over 35 cycles. Upon transitioning to a standing position with normal breathing, the mean heart rate increased to the rate of 84.86 ± 10.39 [BPM] ($n=13$ cycles). Subsequently, during the deep breathing phase while standing, the mean HR was calculated as 97.35 ± 24.16 [BPM] ($n=12$ cycles) according to Eq.[6], exhibiting a notably larger standard deviation compared to the other states. To evaluate whether the change in breathing pattern induced a statistically significant difference in heart rate, a two-sample t-test (Welch's approximation) was performed according to Eq.[5]. The statistical analysis yielded a t-statistic of 1.6552 and a p-value of 0.119 ($df = 14.68$). Since the obtained p-value is greater than the alpha level of 0.05, no statistically significant difference was observed between the heart rates of the tested conditions, despite the visible increase in the mean value during deep breathing.

4.2 Part B: HRV and SNR Statistics

4.2.1 Heart Rate Variability (HRV)

Lead II was selected for the Heart Rate Variability (HRV) analysis due to the superior signal quality and distinct R-peak morphology observed in the previous section. Prior to calculating the heart rate parameters, the raw ECG signal underwent preprocessing, which included bandpass filtering to attenuate noise and baseline correction to ensure a stable isoelectric line. Following R-peak detection, a vector of instantaneous heart rate values was generated for the entire recording duration. The analysis yielded a mean heart rate of 70.71 [BPM] with a standard deviation (STD) of 4.10 [BPM]. To visualize the beat-to-beat variability, a sequence of 15 consecutive heartbeats is presented in Figure 13. The graph displays the instantaneous heart rate for each beat alongside the calculated mean (dashed line) and the upper and lower boundaries defined by ± 1 standard deviation (66.61 [BPM] to 74.80 [BPM]). As illustrated in the figure, while the majority of the data points fluctuate near the mean, several beats (specifically at indices 1, 2, 6 and 15) exceed the upper standard deviation boundary.

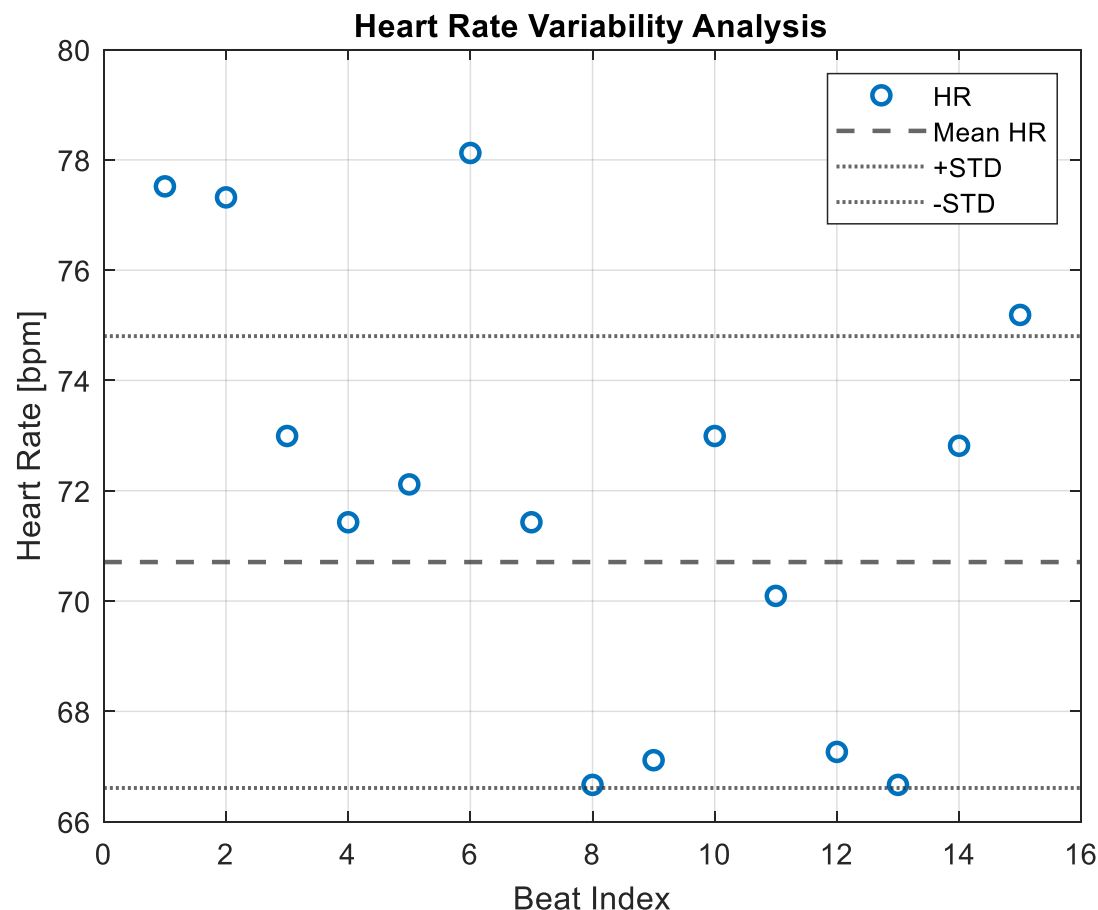


Figure 13: Instantaneous heart rate of 15 consecutive beats, showing the mean and ± 1 standard deviation limits.

4.2.2 Signal Averaging and SNR Improvement

To enhance the signal quality and assess the Signal-to-Noise Ratio (SNR) improvement, the continuous ECG recording was segmented into individual cardiac cycles. R-peaks were identified as the synchronization reference points. A time window of 500 [ms] was selected for each segment, spanning from 200 [ms] prior to the R-peak to 300 [ms] following it, ensuring the inclusion of the complete P-QRS-T complex. A total of 70 beats were initially detected. An outlier rejection algorithm was applied to exclude artifacts or ectopic beats, resulting in the rejection of 2 outliers. Consequently, 68 clean cardiac cycles were utilized to generate the ensemble average.

Figure 14 illustrates the ensemble statistics compared to individual beats. The top panel displays the calculated mean template (Ensemble Mean) bordered by the ± 1 standard deviation boundaries, indicating the variability of the waveform across the recording. The middle and bottom panels present a comparison between the Ensemble Mean and specific single beats (the first valid beat and the last valid beat, respectively). Visual inspection reveals that while the individual beats closely follow the morphological pattern of the mean template, they exhibit higher frequency noise and baseline fluctuations, particularly in the isoelectric segments. In contrast, the averaged signal appears significantly smoother.

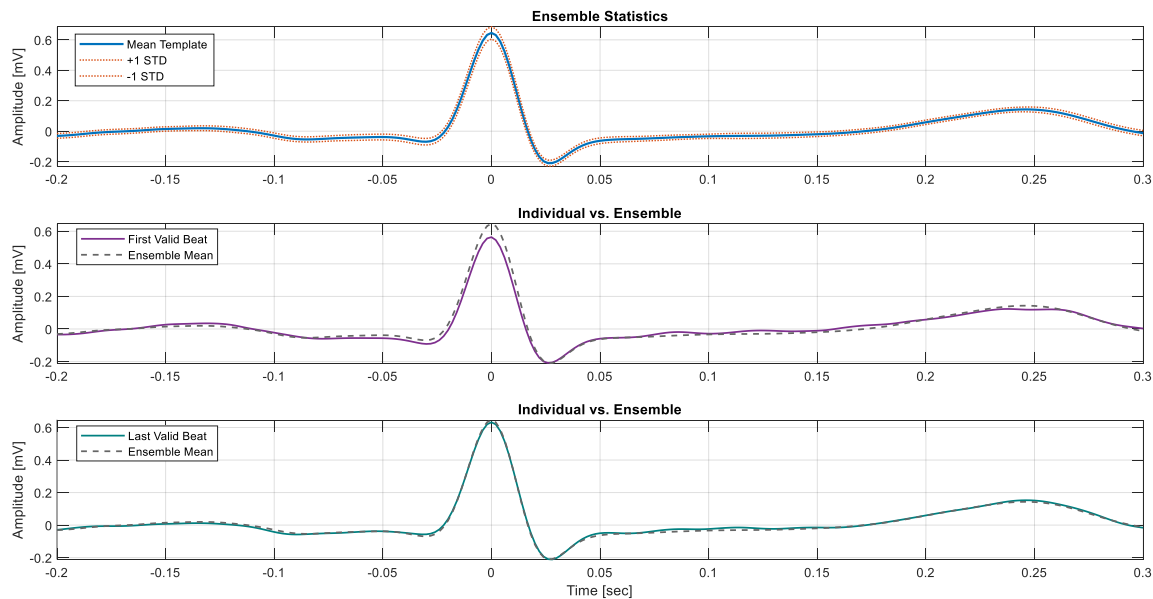


Figure 14: Ensemble statistics of the ECG segments. **Top:** The mean ECG template (blue) with ± 1 standard deviation (orange dotted lines). **Middle:** The first valid beat (purple) superimposed on the ensemble mean. **Bottom:** The last valid beat (teal) superimposed on the ensemble mean.

3.2.3 Quantitative SNR Analysis

To quantify this reduction in noise, the standard deviation (STD) of the signal amplitude was calculated within a 20 [ms] isoelectric segment. The analysis revealed that the noise STD for the first single beat was 0.0102 [mV], whereas for the averaged signal, the noise STD decreased to 0.0095 [mV]. Consequently, the SNR, calculated as the ratio of the R-peak amplitude to the noise STD, Increased from 55.43 in the first single beat to 67.71 in the averaged signal. Figure 15 illustrates this gain, displaying the normalized SNR of the average beat against the first single beat and the difference signal (Δ SNR), which confirms the increased signal prominence relative to the background noise achieved through averaging.

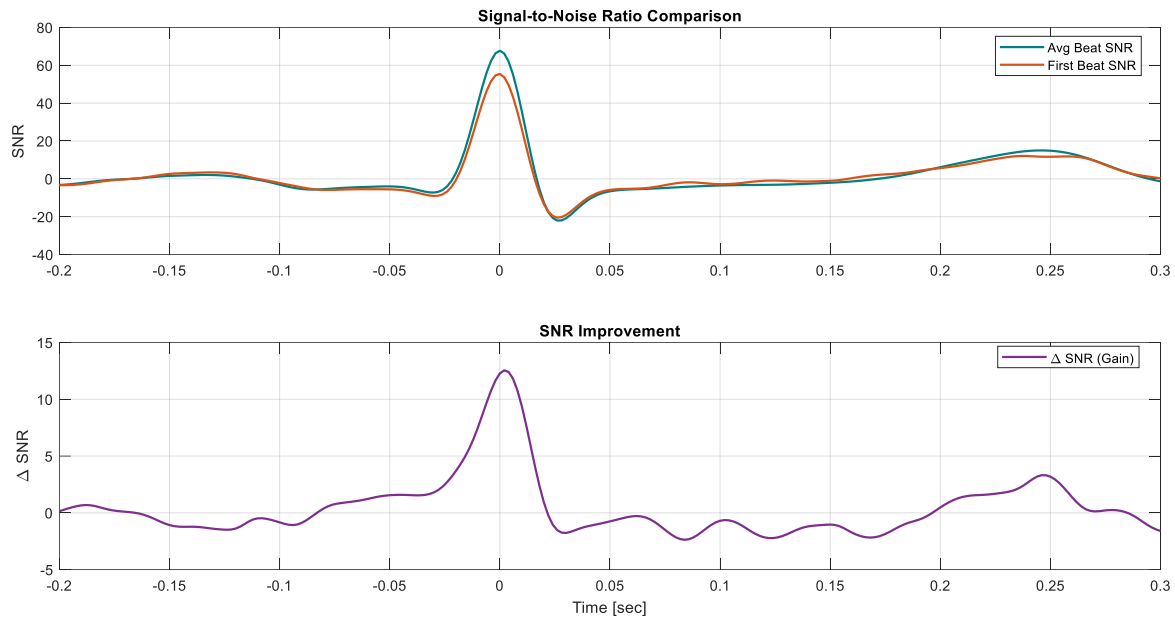


Figure 15: (top panel) shows the normalized "First Beat" and "Avg Beat". The bottom panel, labeled " $\Delta \text{SNR (Gain)}$ ", illustrates the difference between the normalized averaged signal and the normalized single beat. A positive gain is observed around the QRS complex.

4.3 Part C: Heart Sounds Analysis

An automated algorithm was implemented to identify the first (S1) and second (S2) heart sounds from the recorded phonocardiogram (PCG) signal. To visualize the electromechanical coupling of the heart, the detected sounds were plotted alongside the synchronized ECG recording. Figure 16 displays the identification of S1 and S2 superimposed on the PCG signal during the resting state, aligned with the R-peaks of the ECG. The algorithm generally detected the acoustic events corresponding to the mechanical systole and diastole.

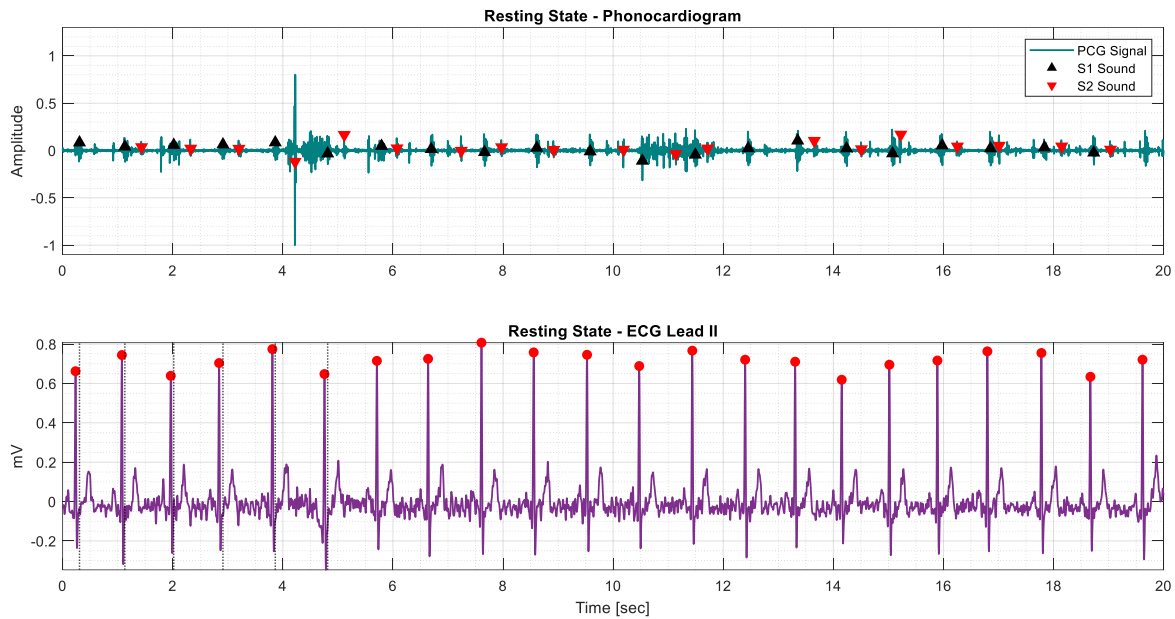


Figure 16: Phonocardiogram (PCG) analysis during the resting state. The top panel shows the PCG signal with detected S1 (black triangles) and S2 (red triangles) sounds. The bottom panel displays the synchronized ECG Lead II signal.

Following physical exertion, the analysis was repeated. Figure 17 illustrates the PCG and ECG signals recorded post-exercise. Visual inspection reveals a significant increase in the amplitude of the heart sounds and a shorter cardiac cycle duration compared to the resting state. The S1 and S2 markers demonstrate the temporal shifts in the mechanical cardiac events relative to the electrical activation.

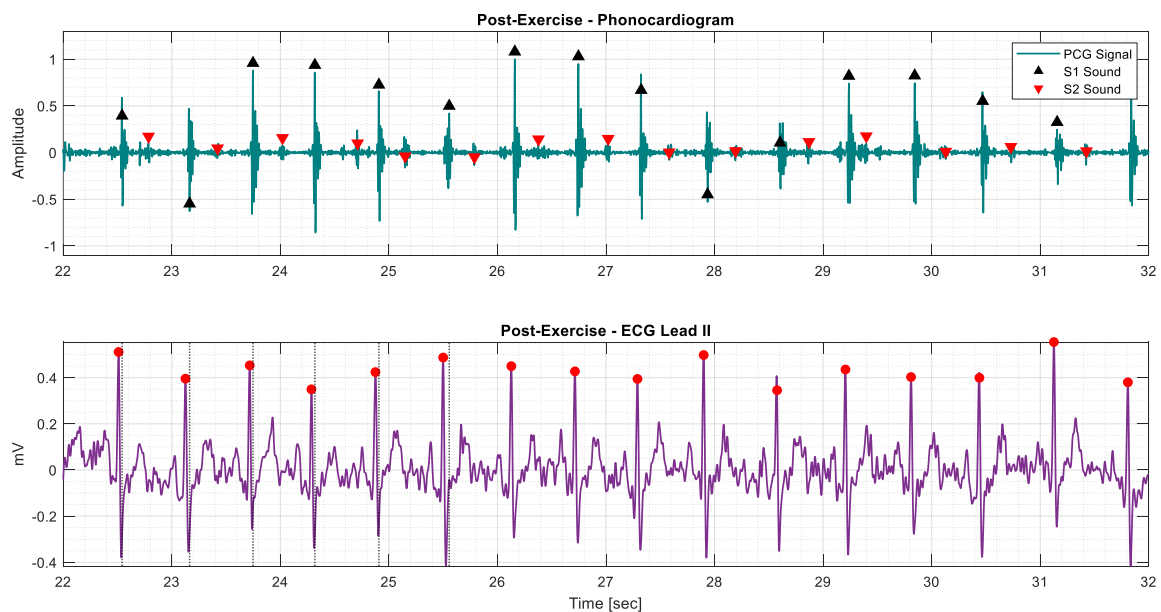


Figure 17: Phonocardiogram (PCG) and ECG signals recorded post-exercise. Note the increased heart rate and the shortening of the systole and diastole intervals compared to the resting state.

The quantitative parameters extracted from the electro-mechanical analysis are summarized in Table 1. The heart rate increased by approximately 49% following exercise. Concurrently, the time intervals

characterizing the electromechanical delay (ΔT R-wave to first sound) and the mechanical systole (ΔT first to second sound) exhibited significant shortening.

Measurement	At Rest (mean $\pm$ std)	Post-exercise (mean $\pm$ std)	Change in % between rest and post-exercise (based on mean values)
BPM [# /min]	65.16 ± 3.20	97.11 ± 5.69	49.05%
ΔT R-wave to 1 st sound [sec]	0.0621 ± 0.0127	0.0341 ± 0.0068	-45.03%
ΔT R-wave to 2 nd sound [sec]	1.6090 ± 0.3833	0.2929 ± 0.0492	-81.80%
ΔT 1 st to 2 nd [sec]	1.5469 ± 0.3800	0.2587 ± 0.0497	-83.28%
ΔT 2 nd sound to next 1 st sound [sec]	-0.5884 ± 0.3687	0.3512 ± 0.0626	-159.70%

Table 1: Electro-mechanical timing intervals of heart sounds and ECG activity at rest and post-exercise.

5 Discussion

The primary objective of this experiment was to record and analyze the electrical and mechanical activity of the heart under various physiological conditions, and to evaluate the effectiveness of digital signal processing techniques in enhancing signal quality. The results demonstrate a clear synchronization between the electrical ECG signal and the mechanical heart sounds, as well as the utility of filtering and signal averaging in improving the Signal-to-Noise Ratio (SNR).

5.1 ECG Signal Analysis and QRS Detection

The analysis of the raw ECG signals revealed the presence of significant noise artifacts, including low-frequency baseline wander and high-frequency interference, which obscured the morphology of the P and T waves. The application of a linear-phase FIR band-pass filter (0.5–50 [Hz]) proved essential for improving ECG signal quality. The filter effectively removed baseline wander and attenuated high-frequency noise while preserving the temporal morphology of the ECG waveform. Maintaining phase linearity was particularly important for this study, as subsequent analyses relied on precise timing information, including R-peak detection, heart rate variability estimation, and synchronization between ECG and phonocardiogram signals.

The DF1 algorithm used for QRS detection demonstrated high sensitivity, successfully identifying all R-peaks across the recorded conditions without false positives or false negatives. However, the algorithm's reliance on fixed thresholds makes it potentially susceptible to errors in the presence of high-amplitude noise or muscle artifacts (EMG) that mimic the spectral characteristics of the QRS complex. A potential improvement to the algorithm would be the implementation of adaptive thresholds that adjust dynamically based on the signal's running average, thereby improving robustness against sudden amplitude changes or wandering baselines.

5.2 Physiological Response and Statistical Significance

The heart rate analysis indicated a physiological response to postural changes, with the mean heart rate increasing from 72.7 [BPM] at rest to 84.9 [BPM] while standing. This elevation is consistent with the baroreceptor reflex, which increases heart rate to maintain blood pressure against gravity. During deep breathing, the mean heart rate further increased to 97.4 [BPM], accompanied by a high standard deviation.

Despite the observed increase in heart rate, the t-test comparing normal and deep breathing yielded a p-value of 0.119, indicating no statistical significance. This lack of significance likely stems from the small sample size ($N=1$) and the short duration of the recording. Physiologically, deep breathing induces Respiratory Sinus Arrhythmia (RSA), where heart rate fluctuates significantly between inspiration and expiration. This natural variability increases the standard deviation, which, combined with a small dataset, reduces statistical power. A longer recording duration or a larger subject pool would likely yield a statistically significant difference.

5.3 HRV and SNR Improvement

The application of signal averaging resulted in a notable improvement in signal quality. The SNR increased from 55.43 in a single beat to 67.71 in the averaged signal. This improvement is attributed to the random nature of the noise, as uncorrelated noise averages toward zero, the time-locked ECG signal is reinforced. Theoretically, the SNR improvement is proportional to the square root of the number of averaged segments ($\sqrt{N}$). In this experiment, 68 cycles were averaged. While increasing the number of cycles would further improve the SNR, the improvement follows a law of diminishing returns, doubling the SNR would require quadrupling the number of beats. Therefore, the number of cycles used represented an optimal balance between processing time and signal enhancement.

5.4 Heart Sounds and Electro-Mechanical Coupling

The simultaneous recording of ECG and PCG confirmed the electro-mechanical coupling of the cardiac cycle. Following physical exercise, the heart rate increased by approximately 49%. This chronotropic response was accompanied by a shortening of the cardiac cycle intervals. Notably, the diastolic interval (S2 to the next S1) shortened more dramatically than the systolic interval, a known physiological mechanism to preserve cardiac output during tachycardia.

The Shannon Energy algorithm successfully detected S1 and S2 sounds in both rest and post-exercise conditions. However, the detection sensitivity is highly dependent on the stethoscope placement and ambient noise levels. Potential sources of error include friction noise or heavy breathing sounds post-exercise, which can create false energy peaks. To improve the algorithm, gating techniques using the R-peak from the ECG as a time reference could be employed to define narrow search windows for S1 and S2, thereby reducing false detections.

5.5 Limitations

The study was limited by the use of a single subject, which precludes the generalization of findings to a broader population. Additionally, the short duration of the recordings (10–30 seconds) limited the ability to perform a comprehensive Heart Rate Variability (HRV) analysis, which typically requires longer datasets to assess autonomic function accurately.

6 Conclusions

In conclusion, this experiment successfully demonstrated the integration of electrical and mechanical cardiac events. The results highlight the critical role of digital signal processing, specifically filtering and averaging, in extracting reliable clinical data from noisy physiological signals. The observed physiological responses to posture and exercise were consistent with established cardiovascular theory, although statistical significance was limited by sample size. Future studies should incorporate larger datasets and utilize multiple ECG leads (12-leads) to enable the diagnosis of more complex pathologies. Furthermore, exploring advanced adaptive algorithms and Machine Learning methods would allow for autonomous improvement of signal detection in noisy environments, further enhancing automated detection capabilities.

7 Bibliography

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